

第一章 氢气

Chapter 1 Hydrogen

第一节 氢气的性质

Section 1 Properties of Hydrogen

一、氢气的物理性质

I. Physical Properties of Hydrogen

在通常状况 (common status) 下, 氢气是一种无色、无味的气体。它的密度非常小, 只有空气的 $\frac{1}{14}$, 是世界上已知的最轻的气体。氢气无毒, 难溶于水。

Under normal conditions, hydrogen is a colorless and odorless gas. Its density is very small, only $\frac{1}{14}$ of that of air, making it the lightest gas known in the world. Hydrogen is non-toxic and sparingly soluble in water.

二、氢气的化学性质

II. Chemical Properties of Hydrogen

氢气在常温下性质稳定 (stable), 但在点燃 (ignite) 或加热条件下, 能够与许多物质发生化学反应。

Hydrogen is stable at room temperature, but under ignition or heating conditions, it can react chemically with many substances.

如图 1.1-1 所示, 纯净的 (pure) 氢气在空气里安静地燃烧, 火焰 (flame) 呈淡蓝色。氢气燃烧生成水, 并放出大量的热。

As shown in Figure 1.1-1, pure hydrogen burns quietly in air, with a pale blue flame. Hydrogen burns to form water and releases a large amount of heat.

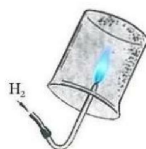
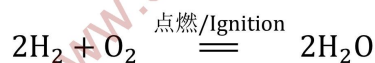


图 1.1-1 氢气的燃烧

Figure 1.1-1 Combustion of Hydrogen

实验测定，空气里如果混入的氢气的体积达到总体积的 4%~74.2%，点燃时就会发生爆炸(explode)。这个范围(range)叫作氢气的爆炸极限(explosive limit)。因此，我们在点燃氢气前，一定要检验氢气的纯度^①。

Experimental determination shows that if the volume of hydrogen mixed in air reaches 4% - 74.2% of the total volume, an explosion will occur upon ignition. This range is called the explosive limit of hydrogen. Therefore, before igniting hydrogen, we must definitely test its purity^①.

【例 1】氢气在空气中燃烧，这个过程是物理变化还是化学变化？

【Example 1】When hydrogen burns in air, is this process a physical change or a chemical change?

解：要判断某一变化是物理变化还是化学变化，核心依据是“是否有新物质生成”——物理变化仅改变物质的状态、形状等，不产生新物质；化学变化会生成不同于反应物的新物质。氢气在空气中燃烧时，反应物是氢气(H_2)和氧气(O_2)，反应后生成了水(H_2O)，水是与氢气、氧气性质完全不同的新物质，因此该过程属于化学变化。

Solution: To determine whether a change is a physical change or a chemical change, the core criterion is "whether new substances are formed" - physical changes only change the state, shape, etc. of substances and do not produce new substances; chemical changes produce new substances different from the reactants. When hydrogen burns in air, the reactants are hydrogen (H_2) and oxygen (O_2), and after the reaction, water (H_2O) is formed. Water is a new substance with completely different properties from hydrogen and oxygen. Therefore, this process belongs to a chemical change.

氢气在加热条件下，能够和氧化铜(CuO)反应生成水，使黑色的氧化铜发生化学反应，变成红色的金属铜(Cu)，如图 1.1-2 所示。

Under heating conditions, hydrogen can react with copper oxide (CuO) to form water, causing the black copper oxide to undergo a chemical reaction and turn into red metallic copper (Cu), as shown in Figure 1.1-2.

^①氢气纯度的检验方法：收集一试管氢气，用拇指堵住试管，将其移近火焰，移开拇指，如果听到尖锐的爆鸣声，就表明氢气不纯，需要再收集，再检验，直到响声很小，听到“噗”的一声才表明氢气已纯净。

^①Method for testing the purity of hydrogen: Collect a test tube of hydrogen, block the test tube with your thumb, move it close to the flame, remove your thumb. If a sharp popping sound is heard, it indicates that the hydrogen is impure and needs to be collected and tested again until the sound is very small and a "pop" sound is heard, which indicates that the hydrogen is pure.



氢气具有还原性，能夺取氧化铜里的氧，使氧化铜失去氧变成铜。氧化铜被氢气还原(reduce)为金属铜：

Hydrogen has reducing properties and can remove oxygen from copper oxide, causing it to lose oxygen and become copper. Copper oxide is reduced by hydrogen to metallic copper:



图 1.1-2 氢气还原氧化铜

Figure 1.1-2 Reduction of Copper Oxide by Hydrogen

【例 2】氢气和氧化铜在加热条件下反应，这个反应的反应物是____，生成物是____。

【Example 2】When hydrogen and copper oxide react under heating conditions, the reactants of this reaction are____, and the products are____.

解:在化学方程式中，等号左边的物质是参与反应的反应物，等号右边的物质是反应生成的生成物，反应条件标注在等号上方或下方。该反应的方程式为 $\text{CuO} + \text{H}_2 \xrightarrow{\Delta} \text{Cu} + \text{H}_2\text{O}$ ，其中等号左边的物质是氧化铜(CuO)和氢气(H_2)，等号右边的物质是铜(Cu)和水(H_2O)，因此反应物是氢气和氧化铜，生成物是铜和水。

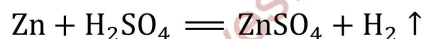
Solution: In a chemical equation, the substances on the left side of the equal sign are the reactants participating in the reaction, and the substances on the right side of the equal sign are the products formed by the reaction. The reaction conditions are marked above or below the equal sign. The equation for this reaction is $\text{CuO} + \text{H}_2 \xrightarrow{\Delta} \text{Cu} + \text{H}_2\text{O}$, where the substances on the left side of the equal sign are copper oxide (CuO) and hydrogen (H_2), and the substances on the right side of the equal sign are copper (Cu) and water (H_2O). Therefore, the reactants are hydrogen and copper oxide, and the products are copper and water.

第二节 氢气的实验室制法

Section 2 Laboratory Preparation of Hydrogen

在实验室(laboratory)里, 我们常用锌(Zn)和稀硫酸(dilute sulphuric acid, H_2SO_4)反应来制取氢气。该反应可以表示为:

In the laboratory, we commonly use zinc (Zn) and dilute sulfuric acid (H_2SO_4) to react to produce hydrogen. This reaction can be expressed as:



由于氢气难溶于水, 因此可以采用排水法收集氢气; 又由于氢气的密度比空气小, 因此也可以采用向下排空气法收集氢气。图 1.2-1 为采用启普发生器(Kipp generator)制备氢气, 同时采用排水法收集氢气; 图 1.2-2 为采用简易启普发生器制备氢气, 同时采用向下排空气法收集氢气。

Since hydrogen is insoluble in water, the drainage method can be used to collect hydrogen; and since the density of hydrogen is smaller than that of air, the downward displacement of air method can also be used to collect hydrogen. Figure 1.2-1 shows the preparation of hydrogen using a Kipp generator and collecting hydrogen by the drainage method; Figure 1.2-2 shows the preparation of hydrogen using a simple Kipp generator and collecting hydrogen by the downward displacement of air method.

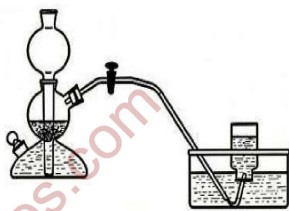


图 1.2-1 启普发生器制备氢气, 排水法收集氢气

Figure 1.2-1 Preparation of hydrogen using a Kipp generator and collecting hydrogen by the drainage method

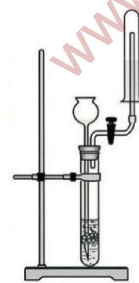
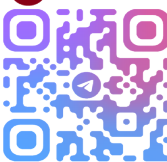


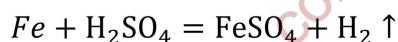
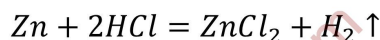
图 1.2-2 简易启普发生器制备氢气, 向下排空气法收集氢气

Figure 1.2-2 Preparation of hydrogen using a simple Kipp generator and collecting hydrogen by the downward displacement of air method



此外(besides), 还可以用盐酸(HCl)代替稀硫酸, 用镁或铁等活泼金属代替锌, 但是不能用不活泼金属(如铜)来制备氢气。相应的反应方程式为:

In addition (besides), hydrochloric acid (HCl) can be used instead of dilute sulfuric acid, and active metals such as magnesium or iron can be used instead of zinc, but inactive metals (such as copper) cannot be used to prepare hydrogen. The corresponding reaction equations are:



像这种由一种单质与一种化合物反应, 生成另一种单质和化合物的反应叫作置换反应(displacement reaction)。

A reaction like this, in which a single substance reacts with a compound to form another single substance and a compound, is called a displacement reaction (displacement reaction).

本章小结

Summary of This Chapter

1. 氢气的性质

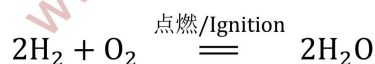
1. Properties of Hydrogen

氢气是一种无色、无味的气体，比空气轻，难溶于水。

Hydrogen is a colorless and odorless gas, lighter than air, and insoluble in water.

氢气能够在空气中燃烧，生成水并放出大量的热。

Hydrogen can burn in the air, producing water and releasing a large amount of heat.



在加热条件下，氢气能够与某些氧化物(如氧化铜)发生氧化还原反应。

Under heating conditions, hydrogen can undergo an oxidation-reduction reaction with certain oxides (such as copper oxide).

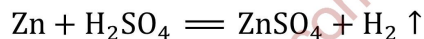


2. 氢气的制备与收集

2. Preparation and Collection of Hydrogen

在实验室里，我们常用锌和稀硫酸反应来制取氢气。

In the laboratory, we commonly use zinc and dilute sulfuric acid to react to produce hydrogen.



氢气可以采用排水法或向下排空气法收集。

Hydrogen can be collected by the water displacement method or the downward air displacement method.